Bootstrap Model Selection in Finite Samples

Resample Size, Screening, and True-Model Selection

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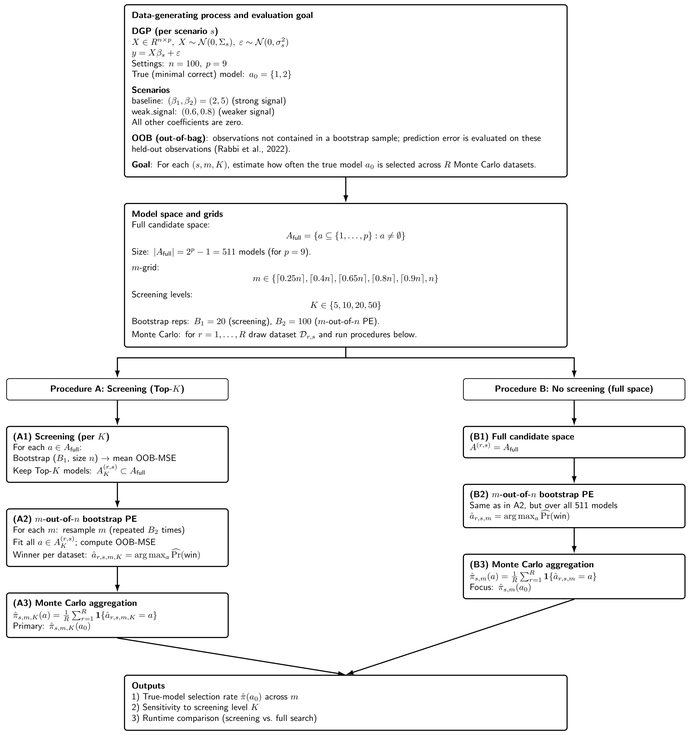
Motivation

- Model selection criteria: Prediction error instead of in-sample fit (Shao, 1996)
- Bootstrap role: Data-driven approximation of prediction error and model comparison (Efron, 1983)
- Standard bootstrap ($m = n$) limitation: Asymptotic inconsistency for smallest correct model (Shao, 1996)
- Model selection: Finite-sample problem → stability perspective of model dominance under resampling (Meinshausen & Bühlmann, 2010)
- Implicit complexity controls: Training size m and candidate set size K (Bickel & Sakov, 2008)
- Research focus: Joint impact of m and K on selection stability, overfitting, and underfitting (Bickel & Sakov, 2008)

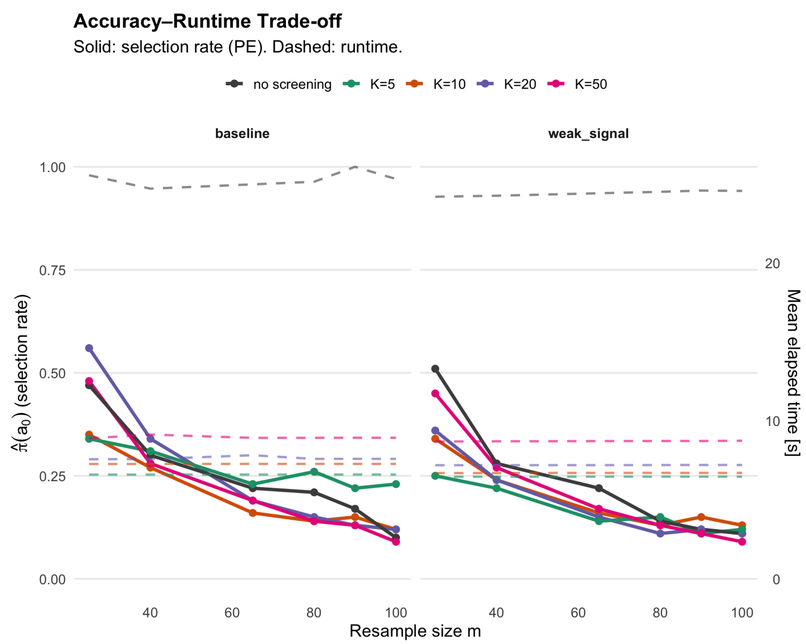
Simulation Design



Methodological Trade-offs

- Choice of K (screening level) (Spokoiny & Willrich, 2019)
  - Smaller K yields large computational gains.
  - Risk: Top-K screening may discard the true model a_0.
  - Screening is noisy (finite B_1): aggressive screening increases exclusion risk.
- Choice of m (m-out-of-n bootstrap) (Shao, 1996; Bickel & Sakov, 2008)
  - $m = n$: classical bootstrap; asymptotically inconsistent for selecting the smallest correct model.
  - Smaller m strengthens the implicit complexity penalty (complex models are unstable when trained on smaller samples).
  - Too small m inflates estimation variance and makes PE estimates noisy.
- Finite-sample tension (Shao, 1996)
  - Larger m: higher stability / lower variance.
  - Smaller m: stronger discrimination among overfitted but correct models.
  - Best-performing m is problem-dependent (signal strength, effective model space).

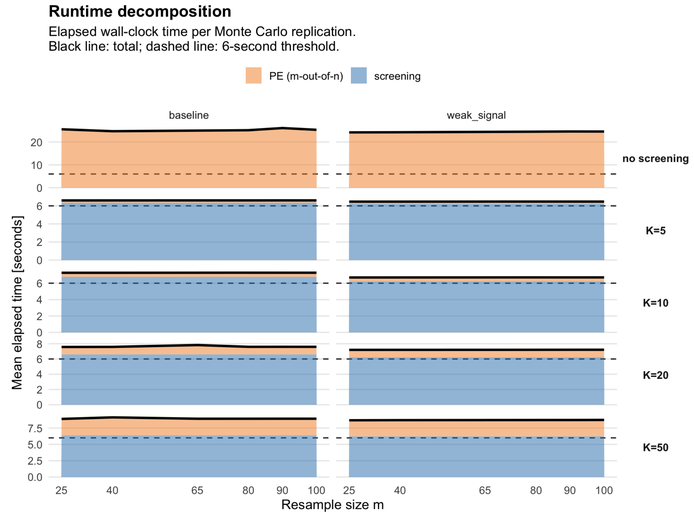
Accuracy–Runtime Trade-off of Bootstrap Model Selection



Solid lines: true-model selection rate $\hat{\pi}(a_0)$ under m-out-of-n bootstrap selection.
Dashed lines: mean runtime per Monte Carlo replication (right axis)
Smaller m can increase discrimination (often higher $\hat{\pi}(a_0)$ in our settings), but too small m may increase noise

Screening trade-off: reduced runtime vs. lower selection accuracy

Runtime Decomposition Across Screening Levels



Discussion and Limitations

- **Empirical frontier.** Across our settings, screening shifts the accuracy–runtime frontier mainly via large runtime gains (see runtime decomposition), at the potential cost of lower $\hat{\pi}(a_0)$.
- **Practical tuning remains open.** While K and m control the trade-offs (see *Methodological Trade-offs*), an *ex ante* choice is difficult in real data because effective signal, sparsity, and redundancy are unknown.
- **Scope.** Finite-sample results for a linear Gaussian DGP and an enumerated model space (moderate p); we illustrate trade-offs rather than establish optimality.

References

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Monte Carlo Simulation Methods

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